Practice Problems: AP Precalc Units 1 and 2

Name: _____ Date: _____

Unit 1: Polynomial and rational functions

Rates of change

W1. Find the average rate of change of $f(x) = 3x - 7$ on $[2, 6]$.

W2. Find the average rate of change of $f(x) = x^2$ on $[0, 2]$ and on $[2, 4]$. Which is bigger, and what does that say about the graph?

1. Use the table.

x	0	1	2	3	4
$f(x)$	1	4	9	16	25

- (a) Find the change in f over each step of 1.
- (b) Are the rates of change increasing or decreasing? Is f concave up or down?
- (c) What kind of function is it, and how do you know?

2. $f(x) = -x^2 + 6x$. Find the average rate of change on $[0, 2]$, $[2, 4]$ and $[4, 6]$. Describe what's happening to the rate of change.

3. A ball's height is $h(t) = -16t^2 + 64t$ feet after t seconds. Find the average rate of change on $[1, 3]$ and explain what the answer means (and doesn't mean).

4. ★ A function is increasing and concave down on an interval. Describe what the rate of change is doing, and give a real-world example.

Polynomials

W3. Find the zeros of $f(x) = (x-2)(x+5)$.

W4. For $f(x) = -2x^3+x-4$, give the degree, the leading coefficient and the end behavior.

5. $f(x) = (x+2)(x-1)^2(x-3)$.

- (a) List the zeros and their multiplicities. Where does the graph cross, and where does it bounce?
- (b) What is the degree? What is the end behavior?
- (c) Find the y-intercept.

6. Find all zeros, real and complex, of $x^3 - x^2 + 4x - 4$. (Hint: grouping.)

7. ★ A degree-3 polynomial with real coefficients and leading coefficient 1 has zeros 2 and $1 + i$. Find the third zero and write the polynomial in standard form.

8. Is each function even, odd or neither? $f(x) = x^4 - 3x^2,\ g(x) = x^3 + x,\ h(x) = x^3 + 1$

9. Use limit notation to describe the end behavior of $p(x) = -3x^5 + 2x^2 - 1$.

Rational functions

W5. For $f(x) = \dfrac{1}{x-3}$, give the vertical asymptote and the horizontal asymptote.

10. $r(x) = \dfrac{2x^2 - 8}{x^2 - x - 6}$. Find the zeros, holes, vertical asymptotes, horizontal asymptote and y-intercept.

11. $r(x) = \dfrac{x^2 + 1}{x - 1}$. Find the vertical asymptote and the slant asymptote.

12. Find $\lim\limits_{x \to \infty} \dfrac{3x + 1}{x^2 + 2}$ and $\lim\limits_{x \to \infty} \dfrac{6x^2 - x}{2x^2 + 5}$.

13. ★ Solve $\dfrac{x - 2}{x + 1} \geq 0$. (Make a sign chart.)

Equivalent forms

14. Divide: $(2x^3 + 3x^2 - x + 5) \div (x + 2)$.

15. Use the binomial theorem or Pascal's triangle to expand $(x+2)^4$.

Transformations

W6. Describe how $g(x) = f(x-3)+2$ relates to $f(x)$.

16. $f(x) = x^2$ and $g(x) = -2f(x+1) - 4$. Describe each transformation in order, and give the vertex of g.

17. ★ The point $(4, 6)$ is on the graph of f. What point must be on $g(x) = 3f(2x) - 1$?

Modeling

18. ★ A quadratic model passes through $(0, 5)$, $(1, 8)$ and $(2, 15)$. Find it, and use it to predict the output at $x = 5$.

Unit 2: Exponential and logarithmic functions

Sequences

W7. Arithmetic or geometric? Give d or r. (a) 3, 7, 11, 15, ... (b) 2, 6, 18, 54, ...

19. An arithmetic sequence has $a_1 = 5$ and $d = 3$. Write a formula for a_n and find a_{20}.

20. A geometric sequence has $g_1 = 4$ and $r = \frac{1}{2}$. Write a formula for g_n and find g_6.

21. An arithmetic sequence has $a_3 = 11$ and $a_8 = 26$. Find d and a_1.

Linear vs. exponential

22. Is each table linear or exponential? Write the function.

x	0	1	2	3
(a) y	5	10	20	40
(b) y	5	8	11	14

23. Find the exponential function $y = ab^x$ through each pair of points. (a) $(0, 3)$ and $(2, 48)$ (b) ★ $(1, 6)$ and $(3, 24)$

Exponential functions and models

W8. Growth or decay, and by what percent? (a) $5(1.08)^x$ (b) $200(0.75)^x$

24. A town of 2,000 people grows 5% per year. Write a model and predict the population after 10 years. (Calculator)

25. A $24,000 car loses 15% of its value each year. Write a model and find its value after 4 years. (Calculator)

26. Rewrite each.

- (a) 3^{x+2} in the form $a \cdot 3^x$
- (b) 8^x as a power of 2
- (c) $\left(\frac{1}{2}\right)^x$ with base 2
- (d) ★ $f(t) = 100(1.2)^t$ grows yearly. What is its monthly growth factor? (Calculator)

27. $1,000 is invested at 4% for 10 years. Find the balance with continuous compounding and with monthly compounding. (Calculator)

Composition and inverses

W9. $f(x) = 2x + 1$ and $g(x) = x^2$. Find $f(g(3))$ and $g(f(3))$.

28. With the same f and g, find $f(g(x))$ and $g(f(x))$ as expressions. Are they equal?

29. Find the inverse of each. (a) $f(x) = \dfrac{x-4}{3}$ (b) $f(x) = 2^x + 1$

30. Show that $f(x) = x^3 - 2$ and $g(x) = \sqrt[3]{x+2}$ are inverses.

Logarithms

W10. Evaluate without a calculator: $\log_2 8$, $\log 1000$, $\ln e^5$, $\log_3 1$, $\log_5 \frac{1}{25}$

31. (a) Rewrite $2^5 = 32$ as a log. (b) Solve $\log_4 x = 3$ by rewriting it as an exponential.

32. Expand: $\log\left(\dfrac{x^3 y}{z^2}\right)$

33. Condense into one log: $2\ln x - \ln(x+1) + \ln 3$

34. Use change of base to find $\log_3 20$ to three decimals. (Calculator)

35. For $f(x) = \log_2(x-3)+1$, find the domain, vertical asymptote and x-intercept.

Equations and inequalities

W11. Solve $2^x = 32$.

36. Solve $3^{2x-1} = 27$.

37. Solve $5^x = 40$. (Exact, then a decimal.)

38. Solve $4e^{2x} = 20$. (Exact, then a decimal.)

39. ★ Solve $\log_2 x + \log_2(x-2) = 3$. Check for extraneous solutions.

40. Solve $\ln(x+1) = 2$.

41. Solve $2^x > 100$.

42. ★ When will the town from #24 reach 5,000 people?

Modeling and semi-log plots

43. A medicine has a half-life of 6 hours. A patient takes 80 mg. Write a model and find how much is left after 15 hours. (Calculator)

44. ★ On a semi-log plot, data lies along the line $\log y = 0.3x + 1$. Find an exponential model $y = ab^x$.

45. Explain why exponential data looks like a straight line on a semi-log plot.

Answer Key

Unit 1

- **W1.** 3
- **W2.** 2 on $[0, 2]$ and 6 on $[2, 4]$. The rate of change is increasing, so the graph is concave up.
- **1.** (a) 3, 5, 7, 9. (b) Increasing, so concave up. (c) Quadratic: the second differences are constant (all 2). It's $(x+1)^2$.
- **2.** $f(0) = 0$, $f(2) = 8$, $f(4) = 8$, $f(6) = 0$, so the rates are 4, 0, -4. The rate of change is decreasing, so the graph is concave down (it rises, peaks, then falls).
- **3.** $h(1) = 48$, $h(3) = 48$, so the average rate is 0 ft/s. The ball ended at the same height it started on that interval. It doesn't mean the ball stopped: it went up and came back down.
- **4.** The rate of change is positive but getting smaller: still going up, but more slowly. Examples: a plant's height as it matures, or a car's speed as it approaches the speed limit.
- **W3.** $x = 2$ and $x = -5$
- **W4.** Degree 3, leading coefficient -2. Up on the left, down on the right: $x \to -\infty$ gives $f \to \infty$, and $x \to \infty$ gives $f \to -\infty$.
- **5.** (a) -2 (mult. 1, crosses), 1 (mult. 2, bounces), 3 (mult. 1, crosses). (b) Degree 4, leading coefficient +1, so up on both ends. (c) $f(0) = (2)(1)(-3) = -6$
- **6.** $(x-1)(x^2+4)$, so $x = 1, 2i, -2i$
- **7.** The third zero is $1 - i$ (conjugate pair). $(x-2)(x^2-2x+2) = x^3 - 4x^2 + 6x - 4$
- **8.** f even, g odd, h neither
- **9.** $\lim_{x\to\infty} p(x) = -\infty$ and $\lim_{x\to-\infty} p(x) = \infty$
- **W5.** Vertical asymptote $x = 3$, horizontal asymptote $y = 0$
- **10.** $r(x) = \dfrac{2(x-2)(x+2)}{(x-3)(x+2)}$. Zero $x = 2$. Hole at $x = -2$, at height $\frac{8}{5}$: the point $\left(-2, \frac{8}{5}\right)$. Vertical asymptote $x = 3$. Horizontal asymptote $y = 2$. y-intercept $\frac{4}{3}$.
- **11.** Vertical asymptote $x = 1$. Long division gives $x + 1 + \dfrac{2}{x-1}$, so the slant asymptote is $y = x + 1$.
- **12.** 0 and 3
- **13.** $x < -1$ or $x \geq 2$ (-1 is excluded because it makes the denominator 0)
- **14.** $2x^2 - x + 1$ with remainder 3: $2x^2 - x + 1 + \dfrac{3}{x+2}$
- **15.** $x^4 + 8x^3 + 24x^2 + 32x + 16$
- **W6.** Shift right 3 and up 2
- **16.** Shift left 1, stretch vertically by 2, reflect over the x-axis, shift down 4. Vertex $(-1, -4)$, opening down.
- **17.** Need $2x = 4$, so $x = 2$, and $y = 3(6) - 1 = 17$. The point is $(2, 17)$.
- **18.** $c = 5$, $a + b = 3$, $4a + 2b = 10$, so $a = 2$ and $b = 1$: $f(x) = 2x^2 + x + 5$, and $f(5) = 60$.

Unit 2

- **W7.** (a) Arithmetic, $d = 4$. (b) Geometric, $r = 3$.
- **19.** $a_n = 5 + 3(n-1)$, $a_{20} = 62$
- **20.** $g_n = 4\left(\frac{1}{2}\right)^{n-1}$, $g_6 = \frac{1}{8}$
- **21.** $5d = 15$, so $d = 3$ and $a_1 = 5$
- **22.** (a) Exponential (ratios of 2): $y = 5 \cdot 2^x$. (b) Linear (differences of 3): $y = 3x + 5$.
- **23.** (a) $a = 3$, $b^2 = 16$, so $y = 3 \cdot 4^x$. (b) Dividing gives $b^2 = 4$, so $b = 2$, $a = 3$: $y = 3 \cdot 2^x$.
- **W8.** (a) Growth, 8%. (b) Decay, 25%.
- **24.** $P = 2000(1.05)^t$, and $P(10) \approx 3258$ people
- **25.** $V = 24000(0.85)^t$, and $V(4) \approx \$12{,}528.15$
- **26.** (a) $9 \cdot 3^x$. (b) 2^{3x}. (c) 2^{-x}. (d) $1.2^{1/12} \approx 1.0153$, about 1.53% per month.
- **27.** Continuous: $1000e^{0.4} \approx \$1491.82$. Monthly: $1000\left(1 + \frac{0.04}{12}\right)^{120} \approx \$1490.83$.

- **W9.** $f(g(3)) = f(9) = 19$, and $g(f(3)) = g(7) = 49$
- **28.** $f(g(x)) = 2x^2 + 1$ and $g(f(x)) = (2x+1)^2$. Not equal.
- **29.** (a) $f^{-1}(x) = 3x + 4$. (b) $f^{-1}(x) = \log_2(x-1)$.
- **30.** $f(g(x)) = \left(\sqrt[3]{x+2}\right)^3 - 2 = x$ and $g(f(x)) = \sqrt[3]{x^3 - 2 + 2} = x$ ✓
- **W10.** 3, 3, 5, 0, -2
- **31.** (a) $\log_2 32 = 5$. (b) $x = 4^3 = 64$.
- **32.** $3\log x + \log y - 2\log z$
- **33.** $\ln\left(\dfrac{3x^2}{x+1}\right)$
- **34.** $\dfrac{\ln 20}{\ln 3} \approx 2.727$
- **35.** Domain $x > 3$. Vertical asymptote $x = 3$. $\log_2(x-3) = -1$ gives $x - 3 = \frac{1}{2}$, so the x-intercept is $x = 3.5$.
- **W11.** $x = 5$
- **36.** $3^{2x-1} = 3^3$, so $2x - 1 = 3$ and $x = 2$
- **37.** $x = \dfrac{\ln 40}{\ln 5} \approx 2.292$
- **38.** $e^{2x} = 5$, so $x = \dfrac{\ln 5}{2} \approx 0.805$
- **39.** $\log_2[x(x-2)] = 3$, so $x^2 - 2x = 8$, which gives $x = 4$ or $x = -2$. Reject -2 (you can't take the log of a negative). The answer is $x = 4$.
- **40.** $x + 1 = e^2$, so $x = e^2 - 1 \approx 6.389$
- **41.** $x > \log_2 100 \approx 6.644$
- **42.** $1.05^t = 2.5$, so $t = \dfrac{\ln 2.5}{\ln 1.05} \approx 18.8$ years
- **43.** $A = 80\left(\frac{1}{2}\right)^{t/6}$, and $A(15) \approx 14.1$ mg
- **44.** $y = 10^{0.3x+1} = 10 \cdot \left(10^{0.3}\right)^x \approx 10(1.995)^x$, which is about $10 \cdot 2^x$
- **45.** If $y = ab^x$, then $\log y = \log a + x \log b$. That's a line in x, with slope $\log b$ and intercept $\log a$.